Retained-row affine reduction: independent analytical synthesis

DEC-20261102-cd6ac3 · TASK-20261102-74f0a5 · 2026-11-02

Decision: support the conditional theorem, exact diagnostic and source mapping.

The independent review TASK-20261102-5288b1 supplies a complete proof of the graded quotient comparison, the full-field solution bijection, the field-equation deletion identity, and the strict translated k=2 diagnostic. All six assigned joints hold at their specified scope. This decision preserves the counterexamples to stronger claims. It does not admit a new implementation, suite or experiment.

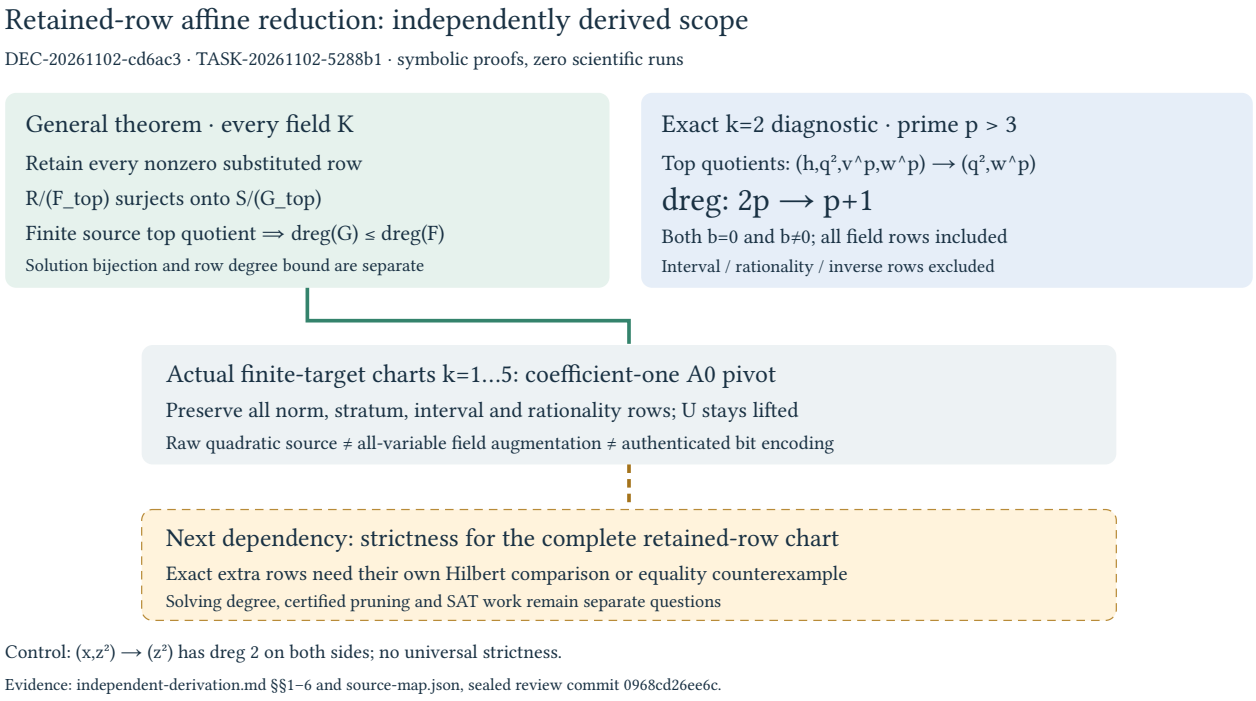


Figure 1: Independent analytical scope. Dashed box denotes the unresolved complete-chart dependency.

Observation — what was independently derived.

Let K be any field, R=K[x,z1,...,zn], S=K[z1,...,zn], n>=0, and L=x-ell(z)-c, with ell homogeneous linear. For F=(L,f1,...,fm), substitute x=ell+c and retain every nonzero resulting row in G. Write I=(F_top) and J=(G_top). Affine evaluation gives the solution map; the separate homogeneous substitution phi(x)=ell gives the graded map. For each degree-d input f, phi(f_top) is the degree-d part of the substituted row: it is its top part if the degree stays d, and zero if the degree drops. Thus phi(I) is contained in J, and

R/I ->> S/J, dim(S/J)_d <= dim(R/I)_d.

If R/I is finite-dimensional, dreg(G)<=dreg(F). Reconstruction x=ell+c is inverse to projection over every extension of K, and individual row degree cannot increase. Nonzero constants are retained; vanished rows can be omitted. The zero quotient has dreg0, while the empty list in no variables has dreg1. These are claims about the highest homogeneous parts of the input list. They are distinct from solving degree, defined through a Macaulay procedure. Definitions1.1 and1.4 in Caminata–Gorla, v2, PDF pages4 and6 (https://arxiv.org/pdf/2112.05579v2) were retrieved by the reviewer and checked by the Coordinator before authoring this synthesis; the paper supplies definitions, not this chart proof.

Over F_p, with ell=sum a_i z_i and c in F_p, the transformed eliminated field equation is exactly sum a_i(z_i^p-z_i). Retaining every surviving field equation, including each auxiliary, certifies its deletion both in the ordinary ideal and in highest parts. All-variable field augmentation bounds monomial exponents by p-1 and proves finite top dimension. Its total input degree is p when variables remain. The raw canonical chart emits quadratic arithmetic without those explicit field equations; authenticated bit/range compilation is a third presentation. Neither finite-field assignment typing nor bit ranges supplies missing polynomial field rows.

Comparison — strict diagnostic with exact rows.

For p>3 prime, a nonsingular curve $y^2=X^3+\alpha X+\beta$ and affine target (a,b), put $c_2=3a$, $c_1=3a^2+\alpha$. In F_p[h,q,v,w], use only

c2-q²-w, c1-2qh-v, b²-h², h-b,
h^p-h, q^p-q, v^p-v, w^p-w.

No interval, rationality, stratum-inverse or other auxiliary equations occur in this diagnostic. Before substitution, the top ideal is (h,q^2,v^p,w^p), with exact basis $q^\varepsilon v^i w^j$, $\varepsilon\in\{0,1\}$, $0\le i,j<p$. Its Hilbert polynomial is $(1+t)(1+\dots+t^{p-1})^2$, with last nonzero degree $2p-1$. After h=b, retain every nonzero row and the remaining field equations. For b=0 the top ideal is (q^2,v,w^p); for $b\neq 0$ it is $(q^2,v+2bq,w^p)$. Both quotients are graded-isomorphic to $F_p[q,w]/(q^2,w^p)$, whose basis is $q^\varepsilon w^j$ and whose last nonzero degree is p. Therefore dreg is exactly 2p before and p+1 after, in both ordinate cases. These are analytical integers, not measured runtime or solving-degree observations; no sampling uncertainty is applicable.

The original source coordinates are r0=chart.A0, r1=chart.A1, s0=support.u0, s1=support.u1. Translation X=a+Z gives h=r0+a r1, q=r1, w=s1+2a, v=s0+a s1+a², with inverse r0=h-aq, r1=q, s1=w-2a, s0=v-aw+a². The translated norm coefficients are the negatives of the three diagnostic arithmetic rows. This is a polynomial-basis translation: the cubic has a Z² term, so it is not a new short-Weierstrass descriptor call. U stays monic and lifted.

Source application — actual finite-target pivot.

The normalized pole basis fixes its highest coefficient to1 and includes an unfixed A0 for finite targets k=1..5. The source sign row A(a)-bB(a) has coefficient+1 on A0. Its affine reconstruction is:

k	A0 reconstructed in original X coordinates
1	-a
2	b-a A1
3	b B0-a A1-a²
4	b B0+ba-a A1-a² A2
5	b B0+ba B1-a A1-a² A2-a³

The normalized constants are included and no division by b occurs. One projected full-field coefficient is removed; all support, inverse, interval and rationality auxiliaries remain. Apply substitution to every norm coefficient and every rationality row. Stratum and interval rows are unchanged; nilpotent/shared-U and constant/shared-B0 rationality bindings are unchanged; guard/slack bindings replace A0 everywhere. Keep all nonzero lower-degree, scalar and redundant rows. The sign row becomes zero. All three arms and every public stratum remain, including impossible combinations with literal obstructions. This is exhaustive family transport through the actual source loop, not renewed validation of imported component code.

There is no sign pivot for identity targets. Identity k0 in the zero stratum is the empty chart with dreg1; the other k0 cases and identity k1 have a literal unit obstruction with dreg0. Identity k2..5 retains its original equations. For finite targets the reconstructed A,B,U and unchanged auxiliaries preserve corresponding full-field coefficient projections and auxiliary fibers, and provide identical inputs to the original signed recovery. Exactly one prescribed contribution at -T remains, including b=0, and both common-root beta signs remain. A differently sized bit projection still requires an authenticated common-output comparison.

Controls — why the scope is necessary.

The independent proof retains seven countercontrols:

- (z²+z,z²,z) and (z²+z,z²) have the same ordinary ideal (z), but dreg1 and2. Ordinary redundancy does not justify row deletion for this metric.
- (x,z²), reduced by x=0, has dreg2 on both sides. Affine elimination is not universally strict, even with redundant prime-field equations.
- With x=z and only x^p-x, dropping the substituted field row leaves an infinite-dimensional top quotient; extension-field solutions also change.
- With x=a z and a outside F_p, the extra (a^p-a)z term invalidates ordinary field-row redundancy; z=1 witnesses it.
- Boolean x=y+z over p>3 needs the transformed x idempotence constraint: y=z=1 satisfies the other bit equations but gives residual2.
- Adding w=0 to the diagnostic changes its regularities to p+1 and2. Its thresholds cannot be copied into a system with additional complete-chart rows.
- Identity charts lack the sign pivot, and literal impossible charts already have the unit ideal. Both defeat indiscriminate strictness claims.

Inference and limitation.

The Coordinator accepts the six proofs and source mapping at these premises. The prior expectation that affine substitution can expose lower-degree information survives the independent challenge. General non-increase is proved conditionally; strictness is proved only for the displayed diagnostic. Complete-chart strictness, solving degree, matrix work, certified pruning and SAT search costs remain unresolved. No subsequent two-pivot, row-pruning, projected-blocking or strictness-criterion argument in `.resume/affine-two-pivot-20261102/` is read into or promoted by this decision.

Custody, independence and procedure.

The reviewer worked in one fresh statement-only session with actual composition/chart source and the permitted primary definitions. Producer proofs, copies, reports and prior review results stayed hidden throughout; there was no blindness lift. The native declared-independence checker passed. It checks the declared inventory and joint ownership, not model identity or mathematical truth. The Coordinator rechecked all four sealed artifact hashes and all14 frozen input hashes, and reconciled the written proofs with the source families. Requested review policy/effort was review-adversarial/xhigh; actual provider, model, tier, effective effort and context/output capacities remain unauthenticated under the approved prospective amendment.

The dispatch list omitted the already-approved reference URL. Root clarified its permission before access. The reviewer retrieved it after initial source inspection and preliminary algebra, before drafting and sealing the proofs; this departs from literature-first ordering and is retained as a procedural deviation. It did not expose producer proofs or change the deadline. Dispatch was18:01:09.329867Z, seal18:20:34.474171Z, elapsed1165.144304s within the3600s limit; one round, no extension. Peak memory was

not measured. Suites, imports of candidate code, CAS, solvers, censuses and remote scientific jobs were all zero. These are documentary/analytical results, with no scientific performance measurements.

Review publication commit: 0968cd26ee6cf1e0bf350b6978310ea5336932d3; root-reported release a1cb35261f and merged PR2981 commit 4f43b0377f1b28bf5a50447a1b9822aa4ffd190a. Root owns this decision's rendering, input manifest, exact review4+ledger6 archive and publication.

Preserved comparison scope.

Keep all106 coordinate cells: F17 curve(2,2), F97 curve(1,4), widths2/3, all affine-x least-positive-y targets, quartic/membership/quadratic/quadratic_aa, complete projected sets, original controls and every phase/failure cost. Keep the distinct six older F17 coefficient cells: degrees8/16, three targets, leaf_sat/prefix_univariate_gcd/exhaustive_reference/aa_reverse_prefix. Neither is replaced by normalized complete-chart coordinates. Keep15 fixed chart fixtures across all three arms and strata, eight nilpotent and eight guard fixtures, mutation/certificate controls, wrapped intervals, targets outside the interval and both beta signs. All five prime-field leads remain in scope; unchanged partial-matrix/two-bit-backdoor scaling remains unapproved. Future comparisons must retain failed models, incomplete/censored work, every attempted call, construction/authentication, compilation, solving, decoding, reconstruction, checking and projection costs.

Next ranked dependency and corpus treatment.

First determine strictness for the actual complete retained-row field-augmented chart, initially finite k2 with all interval/rationality/stratum rows. The required output is an exact graded-ideal/Hilbert comparison or a concrete equality counterexample for b=0 and b!=0, without inheriting the diagnostic thresholds. This directly tests whether the proved presentation change reduces the intended full-chart invariant. Generic engine_bridge.solve authentication, full-cube certificates and complete common signed-output/fiber comparison remain the separate SAT dependency. No new task is admitted here.

The theorem, diagnostic and row-deletion control warrant scoped knowledge curation after this archive. No knowledge entry is created because this handoff permits exactly four source files and no new IDs; the decision records concrete finding/open-question payloads rather than declaring the result unworthy of curation. No novelty claim is made. The prior producer diagram remains an immutable UNREVIEWED snapshot; the full-chart composition diagram remains its historical software-coverage view. The new diagram is the current analytical decision view. Checked frontier rows KR-IC-fbdb61, its successor KR-IC-58213c, and KR-IC-0d2021 describe separate invariant/complexity claims whose premises and bounds are unchanged.

Immutable evidence (internal provenance).

The independent proof and source map carry the detailed argument and source locators. All four artifacts are bound below and in the decision:

- independent-derivation.md (../../../research/prime_sat_affine_regularity_20261102/TASK-20261102-5288b1/independent-derivation.md) — SHA-256 374df66c79c64fc84b976d009b15f3bd50e716d1bd562d31f9a31d96fbf9751a.

- source-map.json (../../../research/prime_sat_affine_regularity_20261102/TASK-20261102-5288b1/source-map.json) — SHA-256 4ae3a6c98c540df393b20c62e54b257e08120f36f958c4e494a1acd2dc983831.

- review.md (../../../research/prime_sat_affine_regularity_20261102/TASK-20261102-5288b1/review.md) — SHA-256 8df217e75127011bc9f10f73b68f62027ff82bbf59ae7439e0b07e70cb191f09.

- attestation.yaml (../../../research/prime_sat_affine_regularity_20261102/TASK-20261102-5288b1/attestation.yaml) — SHA-256 8b9a9979ea2a57237de2219344a6a5a625e84dd118057b15c8cd7d7239fe5645.

Frozen plan: review-plan.yaml (../../../coordination/prime_sat_affine_regularity_20261102/review-plan.yaml). Actual source: composition.py (../../../research/prime_sat_full_chart_composition_20261101/composition.py), SHA-256 354d6eef9695477f5c3ab2e9acc274241848f25a863dac30d2e5eb0def68c3b9; charts.py (../../../research/prime_sat_shared_signed_chart_20261030/charts.py), SHA-256 c808c7d3a5a2a96316a0c13c9faafd848406ef44e960a1bcc52bde700e6a818b.